For Examiner's Use

6 **(a)** Two identical wires A and B, are placed parallel to each other as shown in Fig. 6.1. Both wires carry a current of 90.0 A towards the left.

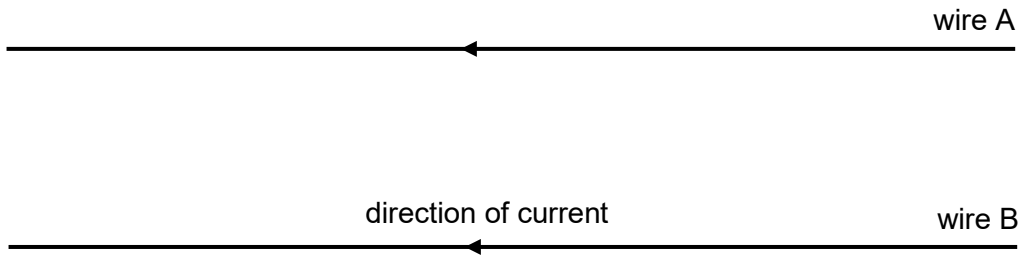

Fig. 6.1

Explain why both wires experience a force.

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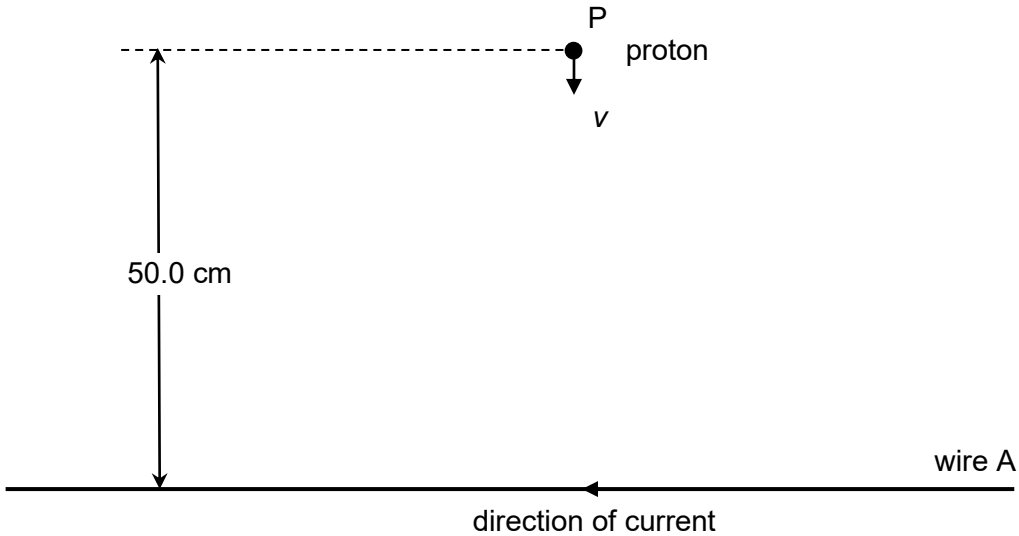
..... [2]

(b) Fig. 6.2 shows only the wire A carrying a current of 90.0 A towards the left. Point P lies in the plane of the paper containing wire A, at a distance 50.0 cm directly above wire A. At P, a proton is travelling directly towards the wire with a speed $v = 1.0 \times 10^3$ m s^{-1}. Ignore the effects of the Earth's magnetic field.



For Examiner's Use

(i) Calculate the radius of the path of the proton when it is at P.

radius = …….....……..... m [3]

(ii) Explain how the path of the proton is affected by the magnetic field produced by the current in wire A as it moves in the region between P and wire A.

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…….....

…… [3]